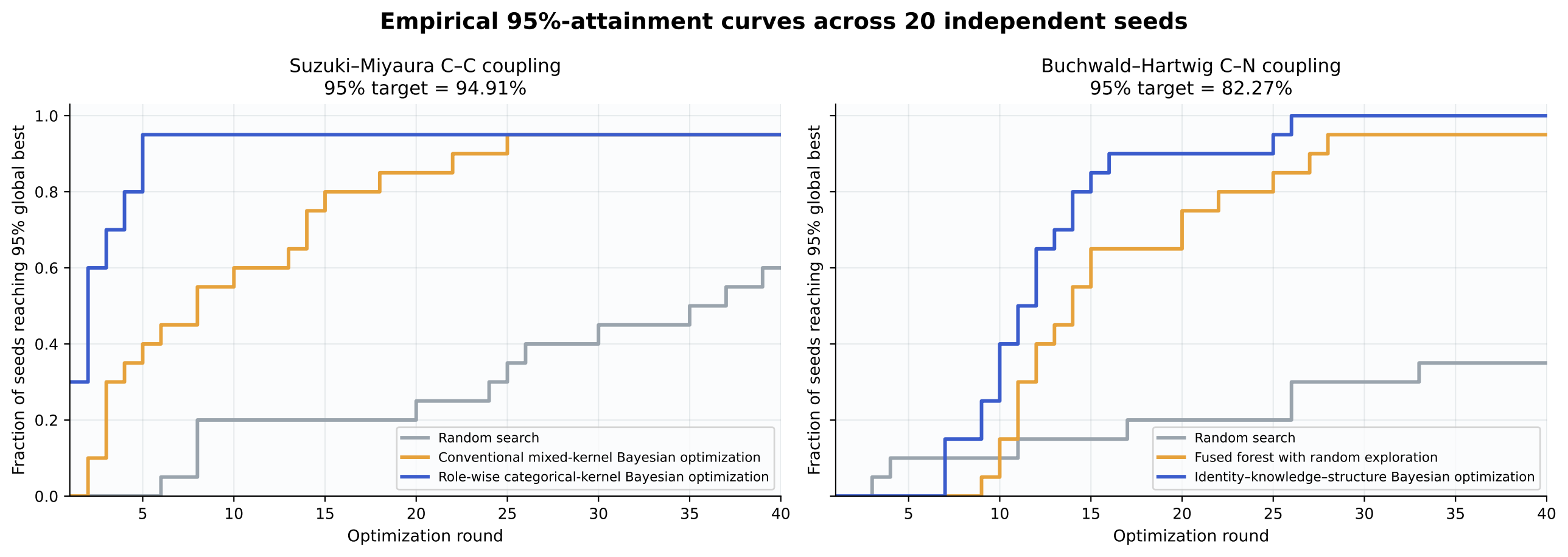
Empirical 95%-attainment curves across 20 independent seeds
Suzuki–Miyaura C–C coupling
95% target = 94.91%
Buchwald–Hartwig C–N coupling
95% target = 82.27%
Fraction of seeds reaching 95% global best
Optimization round
Random search
Conventional mixed-kernel Bayesian optimization
Role-wise categorical-kernel Bayesian optimization
Fused forest with random exploration
Identity–knowledge–structure Bayesian optimization